RAJALAKSHMI ENGINEERING COLLEGE

Department of Computer Science and Design

GE23627 Design Thinking and Innovation

Ideate Phase – Ideation Workshop Sheet

Team Information

Team Name	Student 1 (Roll No and Name)	Student 2 (Roll No and Name)
Team Genzet	241701014 Durgaprasad V	241701049 Sanjaykumar R

Problem Statement:

Citizens face difficulties in reporting and following up on civic issues such as potholes, garbage, drainage problems, water-supply issues, streetlight failures and damaged public facilities. They may not know the responsible authority, have limited visibility into complaint progress, and may not have a reliable way to verify whether a reported issue has actually been resolved. Authorities also face difficulties in identifying duplicate, recurring and unresolved problems across their areas.

Target Users:

- Citizens / Residents
- Elderly and low-digital-literacy citizens
- Persons with disabilities
- Government Department Officers
- Field Workers
- Elected Representatives / Constituency Offices

- Citizen Groups / Resident Associations
- Civic Researchers / Journalists

Brainstorming Rules

1. Focus on generating many ideas.
2. Do not criticize any idea during brainstorming.
3. Encourage creative and unusual ideas.
4. Build on ideas proposed by teammates.
5. Write or sketch ideas

Idea Generation (Brainstorming)

Idea No	Student 1 Idea	Student 2 Idea
1	**Smart Civic Complaint & Reporting App** – A platform where citizens can report potholes, garbage, drainage, water supply, streetlight and other public issues using photos, location, category and description.	**AI-Based Authority Routing System** – Automatically identify the appropriate department or authority for each complaint based on the issue type, location and jurisdiction, reducing delays caused by incorrect routing.
2	**Complaint Tracking & Status Timeline** – Provide citizens with a clear timeline showing complaint submission, verification, assignment, acknowledgement, work progress and resolution status.	**Citizen Resolution Verification** – Allow citizens to confirm whether an issue has actually been resolved through options such as completely resolved, partially resolved or not resolved, with supporting evidence where required.
3	**Recurring Civic Problem Detection** – Analyse repeated complaints in the	**Civic Issue Heatmap & Hotspot Analysis** – Display frequently

	same location or category to identify recurring infrastructure and service problems that may require long-term solutions.	reported civic problems on an interactive map so citizens and authorities can identify areas with repeated or critical issues.
4	**Constituency Accountability Dashboard** – Provide a consolidated view of reported issues, resolved cases, pending cases, overdue complaints, recurring problems and resolution timelines within a constituency.	**Public Project & Expenditure Transparency** – Connect relevant civic issues with verified public development projects, funding sources, approved amounts, expenditure, implementing agencies and project status.
5	**Integrated CivicLens Platform** – Combine citizen reporting, authority routing, complaint tracking, citizen verification, recurring issue analysis, constituency analytics and public project transparency into one civic accountability platform.	**AI Civic Assistant & Infrastructure Memory** – Provide conversational assistance for reporting and tracking civic issues while maintaining a digital history of public assets, repairs, recurring failures and infrastructure-related information.

Idea Voting (Dot Voting)

Idea	Student 1 Vote	Student 2 Vote	Student 3 Vote	Total Votes
Idea 1				
Idea 2				
Idea 3				
Idea 4				
Idea 5				

Top Idea Selected

Selected Idea:

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Concept Development

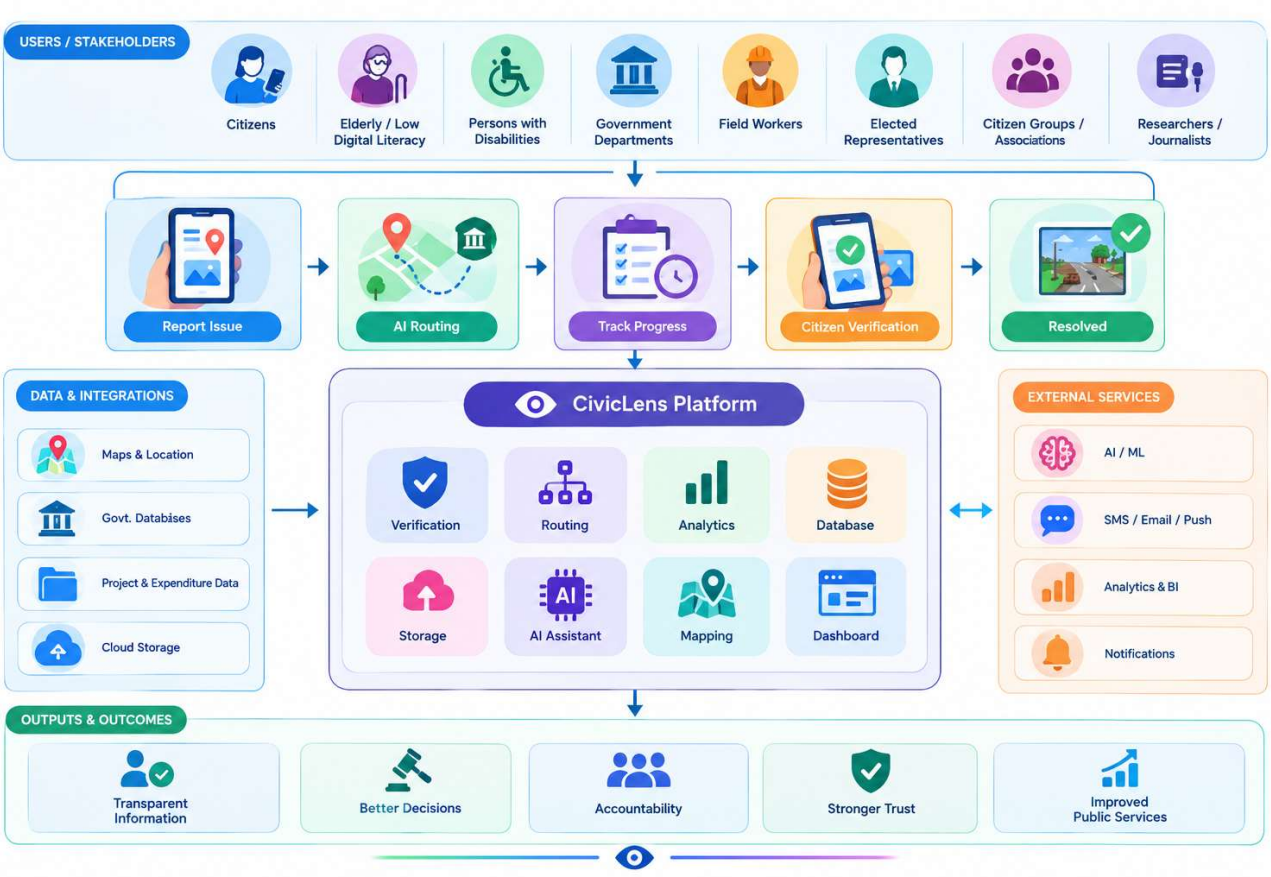
Feature	Description
Project Title	**CivicLens** – Constituency Accountability & Civic Transparency Platform
Key Features	**1. Civic Issue Reporting** • Photo/video evidence • Automatic location detection • Issue category • Text/voice description • Impact information **2. Smart Authority Routing** • Identifies the likely responsible department • Routes complaints to the appropriate authority • Allows authority reassignment when required **3. Complaint Tracking** • Unique complaint ID • Complete status timeline • SLA / expected timeline • Action and progress updates **4. Citizen Verification** • Completely resolved • Partially resolved

	• Not resolved • Evidence-based re-verification • Reopen/escalate unresolved complaints **5. Recurring Issue Intelligence** • Detects repeated complaints • Identifies civic hotspots • Helps authorities identify systemic problems **6. Constituency Dashboard** • Total reported issues • Resolved/in-progress/overdue issues • Critical unresolved issues • Issue categories • Recurring hotspots • Citizen verification rate **7. Public Project & Fund Transparency** • Development projects • Funding source • Approved amount • Expenditure information • Project status • Source and verification information **8. Accessibility** • Tamil + English • Voice-assisted reporting • Simple interface • Large touch targets • Minimal typing • Low-bandwidth support
Technology Required	**Frontend:** • React / Next.js • Tailwind CSS **Backend:** • FastAPI or Node.js / Express

	Database: • PostgreSQL / Supabase **Authentication:** • Supabase Auth / JWT **Storage:** • Supabase Storage **Maps:** • OpenStreetMap-based mapping **Analytics:** • SQL + dashboard layer **Optional AI:** • Issue classification • Authority recommendation • Duplicate complaint detection • Complaint summarization • Language translation • Voice/text extraction • Recurring issue detection **Deployment:** • Vercel + Supabase / backend cloud service
Expected Benefits	**For Citizens** • Easier civic issue reporting • Clear understanding of responsible authority • Transparent complaint progress • Ability to verify actual resolution • Better access to public project information **For Government Departments** • Prioritized complaints • Reduced duplicate reports • Better workload management • Visibility into recurring problems

	• Improved resolution tracking **For Elected Representatives** • Constituency-level overview • Visibility of unresolved and recurring issues • Escalation monitoring • Better oversight without being treated as the executing authority **For the Public** • Greater transparency • Evidence-based civic information • Visibility into public projects and expenditure • Better understanding of civic service delivery

Solution Architecture Diagram



Short Idea Pitch

Problem Addressed:

Citizens often struggle to identify the responsible authority, understand what happens after submitting a civic complaint, track its progress and verify whether the issue has actually been resolved. Authorities also lack a consolidated view of recurring and unresolved civic problems

Proposed Solution:

CivicLens is a citizen-centric civic accountability platform that connects civic issue reporting with authority routing, transparent tracking, citizen verification, recurring-problem analytics and public project/expenditure information.

Key Benefit to Users:

CivicLens makes the entire civic-resolution journey visible, from reporting a problem to verifying its resolution, while connecting civic issues with broader constituency-level patterns and public development information.

Signature of the Coordinator